Formula & Code Base Audit Findings Report

Prepared for Claude Code / Engineering Team • Target: MoneyKeel Calculation Engine

Date:	2026-09-02	**Scope:**	192 Metric Rows & 581 Code Functions
Audit Review:	2026-08-25 Reconciliation	**Status:**	Action Required (High/Medium Priority Remediations)

Executive Summary & Purpose

This document synthesizes the formal formula validation findings, mathematical contradictions, and documentation discrepancies identified across the MoneyKeel Calculation Master spreadsheet. It provides concrete, actionable remediation steps for **Claude Code** to update code implementation, formula specifications, and assumption records.

1. Critical Formula & Mathematical Contradictions

CRITICAL **Performance Returns — Dual Formula Definitions (`MK-XS-M-004` / `MK-PF-F-019`)**

Issue: Discrepancy between mathematical specification and narrative description:
• *Maths Column:* Specifies Modified Dietz: $Return = (Ending - Beginning - NetFlows) / (Beginning + \Sigma(w_i \times Flow_i))$.
• *Words Column:* Describes a holding-weighted average return: $\Sigma(Return_i \times Value_i) / \Sigma(Value_i)$.
• *Code Conflict:* `src/domain/performance/moneyWeighted.ts:80` solves IRR via bisection, whereas `code/engines/window-return.ts:69` calculates Modified Dietz.
Action for Claude Code: Standardize on a single canonical return definition across both engines and sync prose descriptions with the underlying math implementation.

HIGH **Monte Carlo Path & Precision Contradictions (`MK-XS-M-002` / `MK-PL-F-068`)**

Issue: Three opposing specifications on a single metric row:
1. Simulation count: Maths states 1,000 simulations; assumptions state code falls back to 500 when count is omitted.
2. Band display: Prose describes a best-to-worst range; decision notes indicate the band was removed.
3. Rounding: Maths shows exact float; decision notes state nearest 5; assumptions state whole integer.
Action for Claude Code: Clean up fallback logic in `MK-PL-F-068`, standardize Monte Carlo iteration counts, and enforce consistent UI display rounding (e.g., nearest 5 vs exact precision in compare tables).

HIGH **Asset Category Mappings & Aggregation Scope (`MK-XS-M-003` / `MK-NW-F-021`)**

Issue: Category name mismatch impacting account inclusions:
• Math definition lists asset classes as `{brokerage, retirement, other investments}`.
• UI text names classes as *Stocks/Funds, Bonds & CDs, Alternatives, Unclassified*.
• Missing mapping for *Unclassified* determines whether unassigned accounts are included in net worth / investment aggregations.
Action for Claude Code: Explicitly map UI categories to underlying domain asset types in `MK-NW-F-021` to avoid silently dropping or misclassifying accounts.

2. Precision, Rounding & Utility Function Misclassifications

MEDIUM **Rounding Utility Misclassification (`MK-ST-F-055` / `round2`)**

Issue: `MK-ST-F-055` performs `Math.round(n * 100) / 100` (2 decimal places). Audit logs incorrectly report this as "Rounds to a whole number" and "Multiplies by 100 (percentage/options contract)".
Additionally, `MK-ST-F-055` is classified under *ACTION (displays nothing)* despite actively formatting numbers shown to users.
Action for Claude Code: Reclassify `MK-ST-F-055` as a display formatter and update documentation/assumptions across dependent metrics.

3. Key Action Items for Claude Code Pipeline

Component	Target File / Metric ID	Required Code Remediation
Performance Engine	`src/domain/performance/` `MK-XS-M-004`	Resolve conflict between IRR bisection and Modified Dietz. Align function naming so `moneyWeightedReturn` refers strictly to one mathematical algorithm.
Monte Carlo Engine	`MK-PL-F-068` `MK-XS-M-002`	Standardize path fallback defaults (e.g., explicit 1,000 iterations) and normalize rounding rules between summary cards and comparison tables.
Asset Aggregator	`MK-NW-F-021` `MK-XS-M-003`	Implement strict enum mapping between UI account buckets (*Stocks/Funds, Bonds & CDs, Alternatives, Unclassified*) and core engine classification sets.
Utility Helpers	`MK-ST-F-055`	Update function docstrings and sheet audit metadata to accurately reflect 2-decimal rounding scale.